Pythagorean theorem and brace lengths

Day 20 • 2026-12-02 • 2:15 PM–3:30 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 RECOGNISE THE SIDES

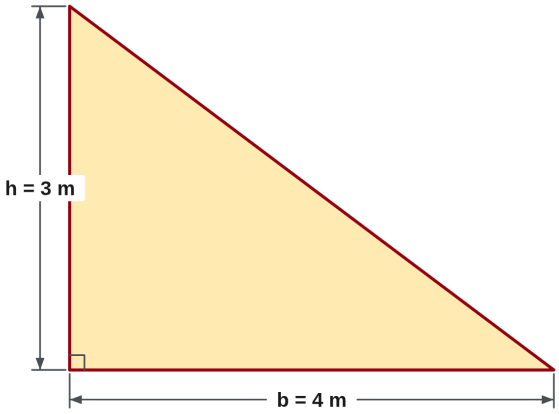
Which side in this Shape Canvas triangle is the hypotenuse?

A. The side opposite the square corner

B. The 3 m leg

C. The 4 m leg

D. Both perpendicular legs



Shape Canvas

Choice (A–D)

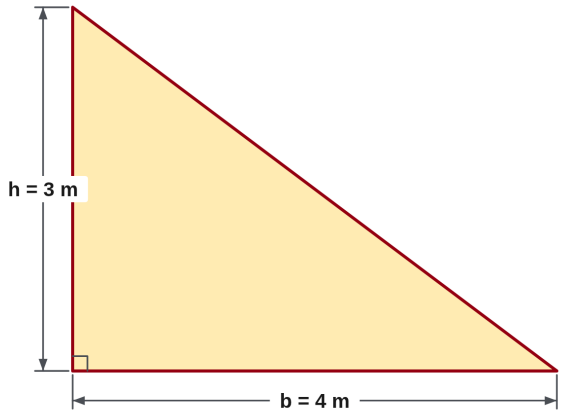
Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

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02 Use the 3 m and 4 m legs shown. What is the sum of their squares?



Shape Canvas

Answer (m²)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

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03 Before checking a brace length with the Numeracy Hub calculator, which expression finds the hypotenuse for 6 m and 8 m legs?

A. $6^2 + 8^2$

B. sqrt($8^2 - 6^2$)

C. sqrt($6^2 + 8^2$)

D. sqrt(6 + 8)



Numeracy Hub

Choice (A–D)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

04 Which condition is required for $a^2 + b^2 = c^2$ in this lesson?

A. The longest side is vertical

B. The triangle has a right angle

C. All three sides are equal

D. The drawing is full size

Choice (A–D)

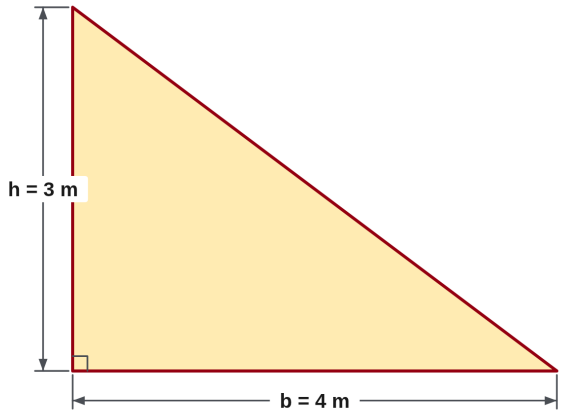
Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

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05 For this Shape Canvas triangle, find the hypotenuse.



Shape Canvas

Answer (m)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

06 FIND A LENGTH

A brace spans perpendicular runs of 5 m and 12 m. Find the brace length.

Answer (m)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

07 A right triangle has a 10 m hypotenuse and one 6 m leg. Find its other leg.

Answer (m)

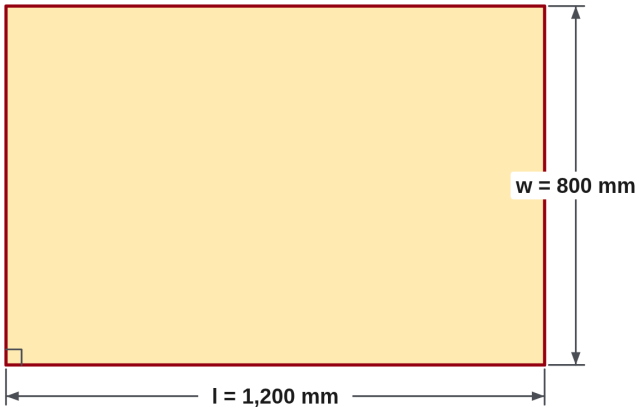
Working / reason

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08 A rectangular panel is 1200 mm by 800 mm as shown. Find its diagonal. Round to 1 decimal place.



Shape Canvas

Answer (mm)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

09 A right triangle has legs 900 mm and 1200 mm. Find its diagonal brace.

Answer (mm)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

10 A 25 m diagonal and a 7 m leg form a right triangle. Find the other leg.

Answer (m)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

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11 VERIFY THE BRACE

A learner reports 7 m as the diagonal of a 3 m by 4 m rectangle. What was likely done?

A. Added the two side lengths instead of using squared lengths

B. Found the correct diagonal

C. Subtracted the side lengths

D. Converted metres to millimetres

Choice (A–D)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

12 Check a 13 m hypotenuse with 5 m and 12 m legs. What is $13^2 - 5^2 - 12^2$?

Answer (m²)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

13 A 6 m and 8 m right-triangle calculation produces 100 m. Which final step was missed?

A. Add 6 and 8

B. Divide by the perimeter

C. Take the square root

D. Multiply by 1000

Choice (A–D)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

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14 A proposed hypotenuse is 4.8 m and the longer leg is 5 m. What should you conclude?

A. Only the units need changing

B. The result needs checking because the hypotenuse cannot be shorter than a leg

C. The result is reasonable

D. The triangle must be equilateral

Choice (A–D)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53

15 A brace is calculated as 2.5 m. Convert this length to millimetres for a cut list.

Answer (mm)

Working / reason

Textbook sections 12.4 pp.301–303; calculator roots pp.51–53